THE REPRESENTING OBJECT OF A TRUNCATED SIMPLICIAL OBJECT IS THE LIMIT OVER THE TRUNCATED SIMPLEX CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-formula-limit (source lines 2553-2570); the proof environment begins two lines below it and is its own proof, closing at source line 2642.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the truncated simplicial objects and the limit involved are developed earlier in the same chapter.

Editorial note. The proof closes with the author's own sentence leaving the verification that the two constructions are mutually inverse bijections and functorial in T to the reader; the retained package reproduces that sentence and does not supply the verification.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: a representing object of the truncated tuple functor is the limit over the truncated simplex category.

Allowed standard prerequisites (editorial).

- n-truncated simplicial object. A sequence of objects U_0, ..., U_n with face and degeneracy maps satisfying the simplicial identities, regarded as a functor on the category of finite non-empty ordinals of size at most n+1.. The face maps and the identities are what the tuple condition in the statement and the verification in the proof use (source lines 2556-2563, 2620-2638).
- Limit over the truncated simplex category. The limit of a truncated simplicial object over the opposite of the category of maps [k] -> [n+1] with k <= n, presented as the set of compatible families.. It is the right-hand side of the statement and the object compared with the representing tuple (source lines 2568, 2573-2580).
- Representable functor. A contravariant functor to sets is representable by an object when there is a natural bijection between its values and the morphisms into that object.. The hypothesis of representability is what lets the proof compare the two functors pointwise on an arbitrary object T (source lines 2565-2566, 2581-2600).

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let n be an integer ≥ 1. Let U be a n-truncated simplicial object of $\mathcal{C}$. Consider the contravariant functor from $\mathcal{C}$ to Sets which associates to an object T the set*

$$\{(f_0, \ldots, f_{n+1}) \in \mathrm{Mor}_{\mathcal{C}}(T, U_n) \mid d^n_{j-1} \circ f_i = d^n_i \circ f_j \ \forall\ 0 \leq i < j \leq n+1\}$$

If this functor is representable by some object U_{n+1} of $\mathcal{C}$, then

$$U_{n+1} = \lim_{(\Delta/[n+1])^{opp}_{\leq n}} U(n)$$

Proof. The limit, if it exists, represents the functor that associates to an object T the set

$$\{(f_\alpha)_{\alpha:[k]\to[n+1],k\leq n} \mid f_{\alpha\circ\psi} = U(\psi) \circ f_\alpha \ \forall\ \psi : [k'] \to [k], \alpha : [k] \to [n+1]\}.$$

In fact we will show this functor is isomorphic to the one displayed in the lemma. The map in one direction is given by the rule

$$(f_\alpha)_\alpha \longmapsto (f_{\delta^{n+1}_0}, \ldots, f_{\delta^{n+1}_{n+1}}).$$

This satisfies the conditions of the lemma because

$$d^n_{j-1} \circ f_{\delta^{n+1}_i} = f_{\delta^{n+1}_i \circ \delta^n_{j-1}} = f_{\delta^{n+1}_j \circ \delta^n_i} = d^n_i \circ f_{\delta^{n+1}_j}$$

by the relations we recalled above the lemma. To construct a map in the other direction we have to associate to a system $(f_0, \ldots, f_{n+1})$ as in the displayed formula of the lemma a system of maps f_α. Let $\alpha : [k] \to [n+1]$ be given. Since $k \leq n$ the map α is not surjective. Hence we can write $\alpha = \delta^{n+1}_i \circ \psi$ for some $0 \leq i \leq n+1$ and some $\psi : [k] \to [n]$. We have no choice but to define

$$f_\alpha = U(\psi) \circ f_i.$$

Of course we have to check that this is independent of the choice of the pair (i, ψ). First, observe that given i there is a unique ψ which works. Second, suppose that (j, ϕ) is another pair. Then $i \neq j$ and we may assume $i < j$. Since both i, j are not in the image of α we may actually write $\alpha = \delta^{n+1}_{i,j} \circ \xi$ and then we see that $\psi = \delta^n_{j-1} \circ \xi$ and $\phi = \delta^n_i \circ \xi$. Thus

$$\begin{aligned} U(\psi) \circ f_i &= U(\delta^n_{j-1} \circ \xi) \circ f_i \\ &= U(\xi) \circ d^n_{j-1} \circ f_i \\ &= U(\xi) \circ d^n_i \circ f_j \\ &= U(\delta^n_i \circ \xi) \circ f_j \\ &= U(\phi) \circ f_j \end{aligned}$$

as desired. We still have to verify that the maps f_α so defined satisfy the rules of a system of maps $(f_\alpha)_\alpha$. To see this suppose that $\psi : [k'] \to [k]$, $\alpha : [k] \to [n+1]$ with $k, k' \leq n$. Set $\alpha' = \alpha \circ \psi$. Choose i not in the image of α. Then clearly i is not in the image of α' also. Write $\alpha = \delta^{n+1}_i \circ \phi$ (we cannot use the letter ψ here because we've already used it). Then obviously $\alpha' = \delta^{n+1}_i \circ \phi \circ \psi$. By construction above we then have

$$U(\psi) \circ f_\alpha = U(\psi) \circ U(\phi) \circ f_i = U(\phi \circ \psi) \circ f_i = f_{\alpha\circ\psi} = f_{\alpha'}$$

as desired. We leave to the reader the pleasant task of verifying that our constructions are mutually inverse bijections, and are functorial in T. □